RESEARCH PROGRESS REPORT · 2 OCTOBER 2026

EdaphNet

Checking soil laboratory reports with an independent sample witness: what has been done, what was found, and what remains

Date: 2 October 2026 · **Project folder:** `edaphnet-v2` · **Repository:** https://github.com/Sudeep-Sharma1/edaphnet-v2 **Purpose of this report:** a complete, plain-language account of what has been done, what was found (including what failed), and what remains. Every number below comes from a committed run record or document, named in brackets.

1. The project in one paragraph

Soil test reports are sometimes wrong: a sample is swapped with another one (the farmer gets someone else's results), or a single value is corrupted (a ×10 decimal slip, a unit mix-up, swapped digits or columns). Nobody checks this for the end user, and re-testing everything is expensive. EdaphNet asks whether a cheap, independent observation of the same soil sample (a **witness**, mainly its colour as a phone camera would see it) can tell a laboratory **which reports to re-test** and **what went wrong**. The core method is a joint model that checks a whole laboratory batch at once, considering possible swaps and a library of error types together.

2. How the research question evolved

Version	Claim	Outcome
v1 (doc 11)	A phone-grade photo lets us flag erroneous reports with a guaranteed false-alarm rate (FDR 10%)	**Failed** (step 5, doc 17), partly because of a calibration-design error of mine (D-016/D-017)
v2 (doc 18, pre-registered)	With a 5% re-test budget, the method finds more errors than the best existing method, with a phone-grade colour witness	**Failed** on new data (KSSL, doc 21)
Current framing	When is a cheap witness enough? Detectability of report errors, calibration requirements, error attribution	Supported by the evidence so far (docs 13, 14, 19, 21)

Both main claims were written down **before** the tests that decided them (pre-registration), and their failures are reported, not hidden.

3. Work completed, step by step

Step	What was done	Output
1. Prior-art search	Papers, patents, products. Most single ideas already exist (phone soil sensing, validating lab values against predictions, swap detection in genomics, cleanlab, conformal FDR). Remaining gap: joint swap + value-error checking with a cheap witness, a detectability map, a benchmark	docs 09, 10
2. Problem specification	Formal model, 9 error types, structured swap patterns, 7 baselines + rule-based repair, 6 hypotheses with pass/fail rules fixed in advance	doc 11
3. Simulation study	Method, baselines and statistics built and tested on synthetic soils (30 settings)	doc 13
4a. Real-data audit	Open Soil Spectral Library downloaded, licences checked, data audited (with a corrected counting mistake)	doc 12
4b. Witness ladder	Soil colour computed from real spectra; measured how much colour reveals about each property	doc 14
5. Go/no-go on real data	ICRAF-ISRIC (3,473 samples), 42 settings	doc 17
5b. Diagnosis	Found the cause of the false-alarm failure (calibration-sample reuse)	D-016, D-017
5c. Pre-registered test on new data	KSSL (19,746 samples), 50 settings, automatic verdict script	docs 18, 21
6 (preparation)	Data-collection protocol, Excel templates, photo-processing module, colour model	doc 15, `templates/`, `phone.py`
8 (start)	Paper skeleton and draft (Introduction, Related work, Method, Data, Protocol)	doc 16, `docs/paper/draft_v0.md`
Demos	Command-line demo, presentation page, live demo with photo + report	`scripts/demo.py`, `demo_site/`

4. What was built

Research code (`src/edaphnet2/`, Python):

Module	Function
`inference.py`	The method: joint batch inference over swap hypotheses × error types; score, explanation, correction
`operators.py`, `swaps.py`	Error library (decimal, unit, digit transposition, column swap, gross error) and swap patterns
`witness.py`, `real_data.py`, `space.py`	Witness models, real-data transforms, colour from spectra, property sets
`baselines.py`	7 comparison methods incl. cleanlab (Zhou et al. 2023)
`conformal.py`, `metrics.py`	False-alarm control (conformal p-values, BH), re-test-budget metric, weighted / per-group calibration
`phone.py`	Photo processing: colour-card correction, true soil colour, blur / exposure / texture
`tracking.py`	Run records (config, code version, environment) for reproducibility

Scripts: experiment runners (synthetic, real, resumable), data builders, analysis (verdicts, detectability), photo processing, colour model training, demos. **Tests:** 23 automatic tests (plus one slow end-to-end resume test), all passing.

5. Data used

Dataset	Samples	Properties	Licence	Role
Synthetic (T0)	generated	8	—	testing the mechanism
ICRAF-ISRIC (OSSL)	3,473 (553 profiles)	pH, OC, clay, sand, CEC, K, Ca, Mg	CC-BY	step 5
KSSL Vis-NIR (OSSL)	19,746 (6,999 locations)	OC, N, total C	CC-BY	pre-registered test
LUCAS	~40,000	10 properties	needs access request	not used yet

6. Key results

6.1 Simulation (doc 13, synthetic)

The method beats all baselines **only when the witness is moderately informative** (e.g. +20 to +25 points in errors found at 10% false alarms when colour explains ~66% of organic carbon), not with a weak witness. Its gain comes almost entirely from **finding swaps**.

6.2 What colour reveals on real soils (docs 14, 21)

	Organic C	Nitrogen	pH	Clay / sand / CEC / K
ICRAF colour (perfect / good phone)	0.37 / 0.35	—	0.27 / 0.15	≤ 0.16 / ≤ 0.07
KSSL colour (perfect / good phone)	0.72 / 0.70	0.55 / 0.54	—	—
Spectrometer	0.60–0.90	0.73	0.62	0.39–0.66

(R^2, share of variation explained on unseen locations.) Colour mainly knows **carbon**.

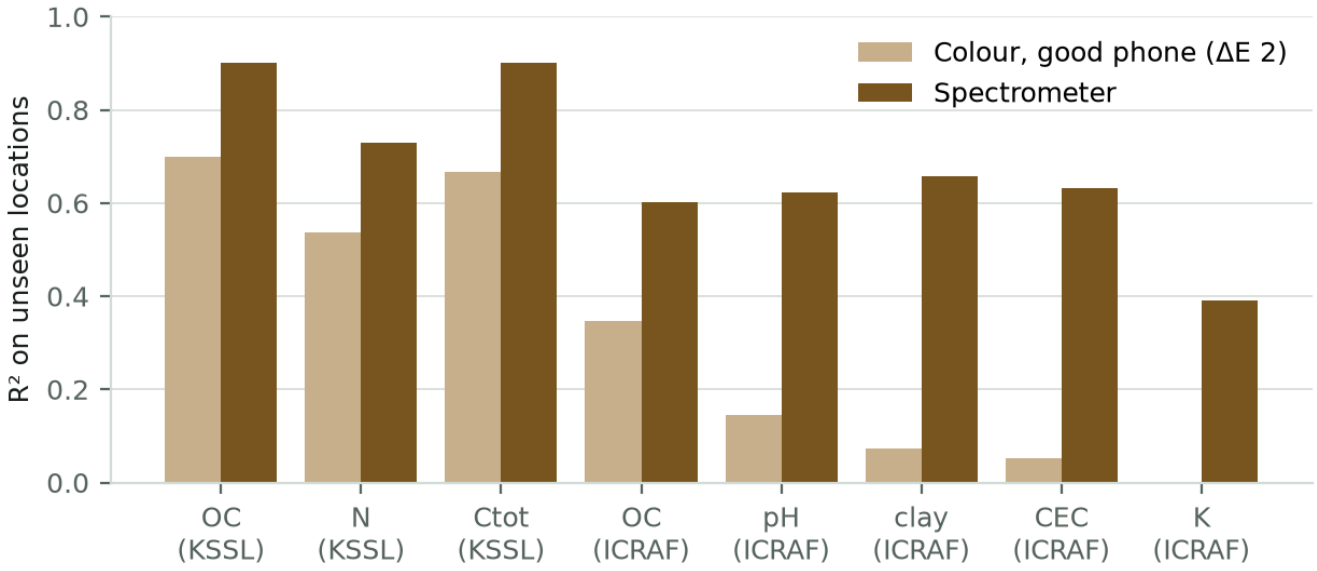


Figure 1. How much each witness reveals about each soil property (R^2, unseen locations). Colour mainly knows carbon; a spectrometer knows much more. Source: witness-ladder run records.

6.3 Pre-registered test on KSSL (doc 21): errors found when re-testing the top 5%

Witness	EdaphNet	Best existing method	Difference [95% CI]
None	0.24	0.27	−0.03
Colour, perfect	0.42	0.42	**+0.00 [−0.02, 0.02]**
Colour, good phone	0.41	0.40	**+0.01 [−0.01, 0.04]**
Spectrometer	**0.60**	0.51	**+0.09 [0.07, 0.11]**

Verdict: A1 (main claim) **failed**; A1-c, A2 (false alarms ≤ 12%) and A3 (attribution) **passed**. Note on A2: with a colour witness the method flagged almost nothing (power ≈ 0), so A2 passed by staying silent at those settings; only the spectrometer setting combined low false alarms with real power.

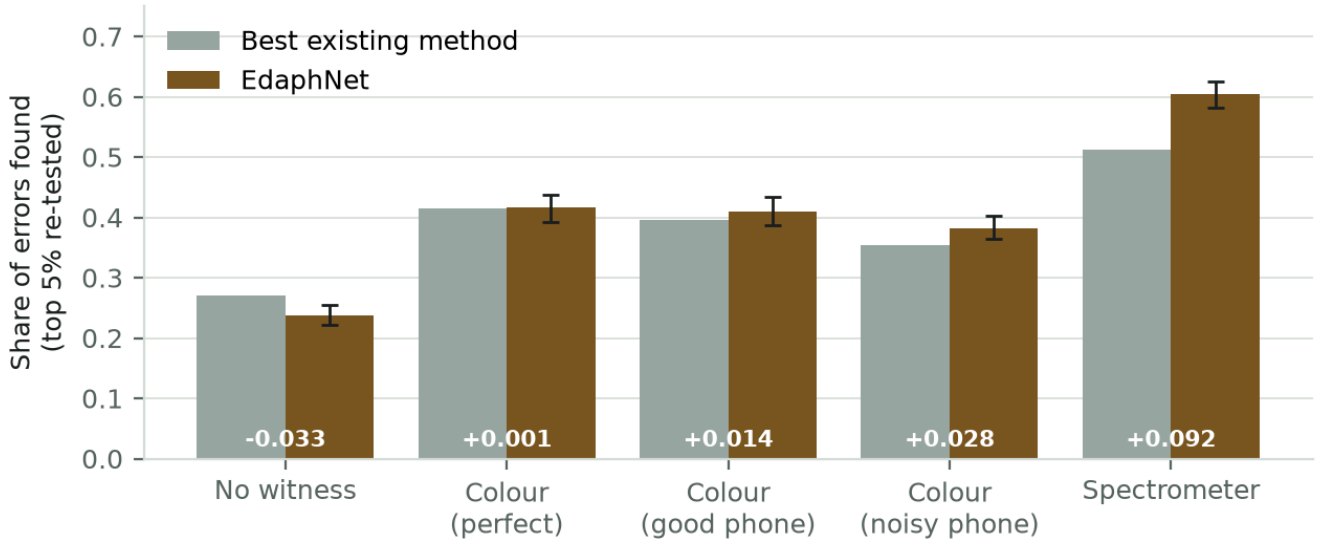


Figure 2. Pre-registered KSSL test: share of erroneous reports found when the top 5% are re-tested (random batches, 5 seeds). Numbers at the bottom: EdaphNet minus best existing method; whiskers: 95% interval of that difference. Source: 20261001-234902_real_kssl3_095deaa.

6.4 Other findings

- **Error explanation:** right error type 89–98% (ICRAF) and 97–100% (KSSL, spectrometer), vs 24–51% for rule-based repair.
- **Which errors can be found (doc 19):** with a colour witness, swaps go from ~3% to ~40–55% found; single wrong values are found better by simple checks; digit transpositions and moderate analytical errors stay hard for every method.
- **Calibration lesson (D-017):** reliable false-alarm control needs **thousands of distinct verified reports**; reusing calibration samples silently breaks it.

7. Mistakes made and corrected (kept on record)

Mistake	Correction
Data audit counted empty spectrum rows (23,852 instead of 49 usable)	Recounted; switched to ICRAF-ISRIC (doc 12 §7)
"Weak witness is enough" (H2)	Refuted by simulation; claim reframed (D-010)
Experiment runner ran out of memory; seeds not reproducible	Lean runner, stable seeds, resume feature with equivalence test (D-012)
Calibration reused samples (pseudo-replication), inflating false alarms	Diagnosed on data not used for testing; fixed by one score per sample (D-016, D-017)

8. Demos and frontend

Demo	How to run	Shows
Command line	`python scripts/demo.py` (`--witness W3`)	one batch, planted swap + ×10 error, re-test list
Presentation page	open `demo_site/index.html`	problem, method, interactive batches, charts, verdicts (works offline)
Live demo	`python scripts/serve_demo.py` → http://localhost:8000	photo → colour correction → prediction → report check → batch ranking, running the real model

The example photos in the demos are **simulated** from real soils' lab-measured colours; no real photos have been collected yet.

9. Reproducibility

Every experiment writes a run record (configuration, code version, environment, results). Seeds are fixed and runs can be resumed with identical results. The pass/fail rules and the verdict script were committed before the data they judge. Data licences and attribution are in the README.

10. What remains

Needs decisions or actions from you / your supervisor

1. Supervisor review of the plan and the new framing (docs 11, 18, 21).
2. LUCAS access request (free): https://esdac.jrc.ec.europa.eu/content/lucas2015-topsoil-data
3. Formal literature search in Scopus / Web of Science (log in `docs/search_log.csv`).
4. Step-6 decisions: budget, soil types and regions, laboratory, photo protocol (doc 15 §11).

Research work

5. Combined re-test policy (EdaphNet for swaps + a value checker), pre-registered and tested on new data (draft: doc 20).
6. Real phone photos + lab reports of Indian soils with blind duplicates (doc 15): the only way to test real phones, which also capture texture.
7. LUCAS confirmation run.
8. Paper: Results, Discussion, Abstract; journal choice.

Engineering (only if the research supports it)

9. A web backend and app for laboratories (step 7).

11. Honest bottom line

The method works well when the witness is strong (spectrometer): it finds more errors than existing methods and explains them correctly. A phone-grade colour photo, at least as approximated from laboratory spectra, is **not enough** to beat the best existing method on average. The publishable contribution is therefore the careful,

pre-registered study of **when a cheap witness is enough**, together with the calibration lesson and the explanation accuracy, unless real phone photos (step 6) or the combined policy change that picture.